1.1 - Systems Of Linear Equations

Notes: The key exercises are 7 (or 11 or 12), 23–26, and 35. For brevity, the symbols R1, R2,…, stand for row 1 (or equation 1), row 2 (or equation 2), and so on. Additional notes are at the end of the section. In Exercises 15–18, students are asked to check their answers to Exercises 11–14; checking that solutions are correct, or at least reasonable, is an important skill in the high-tech industry.

3. The point of intersection satisfies the system of two linear equations:

$$\begin{aligned} x_1 + 5x_2 &= 7 \\ x_1 - 2x_2 &= -2 \end{aligned} \qquad \begin{bmatrix} 1 & 5 & 7 \\ 1 & -2 & -2 \end{bmatrix}$$

Replace R2 by R2 + (–1)R1 and obtain:

$$\begin{aligned} x_1 + 5x_2 &= 7 \\ -7x_2 &= -9 \end{aligned} \qquad \begin{bmatrix} 1 & 5 & 7 \\ 0 & -7 & -9 \end{bmatrix}$$

Scale R2 by –1/7:

$$\begin{aligned} x_1 + 5x_2 &= 7 \\ x_2 &= 9/7 \end{aligned} \qquad \begin{bmatrix} 1 & 5 & 7 \\ 0 & 1 & 9/7 \end{bmatrix}$$

Replace R1 by R1 + (–5)R2:

$$\begin{aligned} x_1 &= 4/7 \\ x_2 &= 9/7 \end{aligned} \qquad \begin{bmatrix} 1 & 0 & 4/7 \\ 0 & 1 & 9/7 \end{bmatrix}$$

The point of intersection is $(x_1, x_2) = (4/7, 9/7)$.

7. Ordinarily, the next step would be to interchange R3 and R4, to put a 1 in the third row and third column. But in this case, the third row of the augmented matrix corresponds to the equation $0x_1 + 0x_2 + 0x_3 = 1$, or simply, 0 = 1. A system containing this condition has no solution. Further row operations are unnecessary once an equation such as 0 = 1 is evident.
The solution set is empty.

9. The system has already been reduced to triangular form. Notice that in the last row, the only nonzero entry is in the last column. The corresponding system of equations does not have a solution. The solution set is the empty set.

11. First, swap R1 and R2. Then replace R3 by R3 + (–3)R1. To reach triangular form, replace R3 by R3 + (2)R2. Next, divide the last row by 4.

$$\begin{bmatrix} 0 & 1 & 4 & -4 \\ 1 & 3 & 3 & -2 \\ 3 & 7 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 & -2 \\ 0 & 1 & 4 & -4 \\ 3 & 7 & 5 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 & -2 \\ 0 & 1 & 4 & -4 \\ 0 & -2 & -4 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 & -2 \\ 0 & 1 & 4 & -4 \\ 0 & 0 & 4 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 & -2 \\ 0 & 1 & 4 & -4 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

To continue, subtract (4)R3 from R2 and subtract (3)R3 from R1. Finally, replace R1 by R1+(–3)R2.

$$\begin{bmatrix} 1 & 3 & 0 & -5 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 19 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

The solution set contains one solution: (19, –8, 1).

15. The solution found in Exercise 11 is (19, –8, 1). Substituting into the equations $\begin{aligned} x_2 + 4x_3 &= -4 \\ x_1 + 3x_2 + 3x_3 &= -2 \\ 3x_1 + 7x_2 + 5x_3 &= 6 \end{aligned}$ results in $\begin{aligned} (-8) + 4(1) &= -4 \\ 19 + 3(-8) + 3(1) &= -2 \\ 3(19) + 7(-8) + 5(1) &= 6 \end{aligned}$, which establishes that the solution found to Exercise 11 is indeed correct.

19. First, replace R4 by R4 + (–3)R1, then replace R3 by R3 + (2)R2, and finally replace R4 by R4 + (3)R3.

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 3 & 0 & 0 & 7 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 0 & 0 & -9 & 7 & -11 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & 0 & 3 & -4 & 7 \\ 0 & 0 & -9 & 7 & -11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & 0 & 3 & -4 & 7 \\ 0 & 0 & 0 & -5 & 10 \end{bmatrix}$$

The resulting triangular system indicates that a solution exists. In fact, using the argument from Example 2, one can see that the solution is unique.

21. Row reduce the augmented matrix corresponding to the given system of three equations:

$$\begin{bmatrix} 1 & -4 & 1 \\ 2 & -1 & -3 \\ -1 & -3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 1 \\ 0 & 7 & -5 \\ 0 & -7 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & -4 & 1 \\ 0 & 7 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

The system is consistent, and using the argument from Example 2, there is only one solution. So the three lines have only one point in common.

23. $\begin{bmatrix} 1 & h & 4 \\ 3 & 6 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & h & 4 \\ 0 & 6-3h & -4 \end{bmatrix}$ Write c for $6 - 3h$. If $c = 0$, that is, if $h = 2$, then the system has no solution, because 0 cannot equal –4. Otherwise, when $h \neq 2$, the system has a solution.

25. $\begin{bmatrix} 1 & 3 & -2 \\ -4 & h & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & h+12 & 0 \end{bmatrix}$. Write c for $h + 12$. Then the second equation $cx_2 = 0$ has a solution for every value of c. So the system is consistent for all h.